TRIDANT Trento Integrated Dielectric Automotive Antenna

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***Abstract*—This work presents a dual-band cylindrical dielectric resonator antenna (CDRA) for combined Global Navigation Satellite System (GNSS) and Vehicle-to-Everything (V2X) applications. The proposed antenna simultaneously provides broadside circular polarization at the GNSS L1 band and quasi-omnidirectional horizon coverage in the V2X band through the excitation of distinct resonant modes within a single dielectric structure. Circular polarization is achieved by means of a custom quadrature hybrid coupler feeding two orthogonal probes, while a conical air cavity is introduced to properly tune the higher-order mode toward the V2X operating frequency. Simulated and measured results demonstrate satisfactory agreement in terms of impedance matching and radiation characteristics. Additionally, a compact low-cost experimental platform for real-time antenna characterization is presented and validated through software-defined radio (SDR)-based polarization measurements.**

***Index Terms*—Dielectric resonator antenna (DRA), cylindrical dielectric resonator antenna (CDRA), GNSS antennas, V2X communications, circular polarization, multimode antennas, software-defined radio (SDR), radiation pattern measurements.**

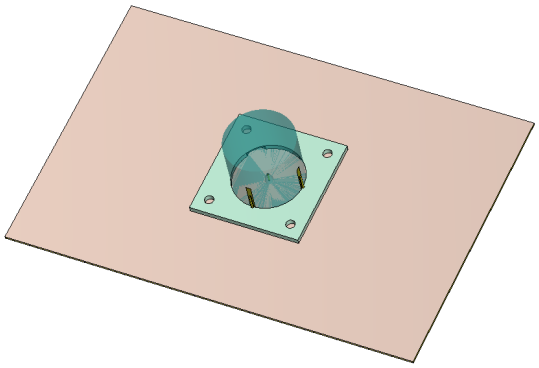
I. Introduction

MODERN automotive platforms increasingly integrate multiple RF services within compact antenna modules, typically housed in roof-mounted structures. Among the most critical are Global Navigation Satellite System (GNSS) reception, which requires reliable upward coverage and circular polarization, and Vehicle-to-Everything (V2X) communication, which demands stable radiation along the horizon to support inter-vehicular and infrastructure links. Developing separate antennas for each function increases installation complexity, cost, and volume, motivating the need for multifunctional antenna solutions.

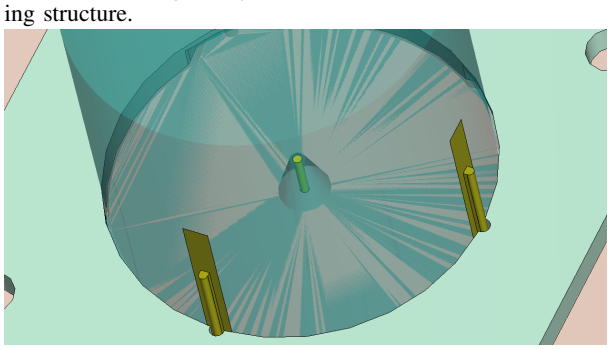
Dielectric Resonator Antennas (DRAs) inherently support multiple electromagnetic eigenmodes, each associated with different resonant frequencies and characteristic radiation behavior. This property makes them strong candidates for multifunctional operation without mechanical reconfiguration. By properly selecting geometry and excitation, it becomes possible to selectively activate different modes and therefore different radiation functions, allowing a compact antenna to simultaneously address distinct automotive communication needs.

This work aims to design, prototype, and experimentally demonstrate a dual-band DRA capable of addressing both GNSS and V2X requirements using a single resonant structure. The objectives are:

- O1: Realize broadside circularly polarized radiation at GNSS L1.



(a) CST model of the proposed dual-band dielectric resonator antenna, including the cylindrical resonator and coaxial feeding structure.



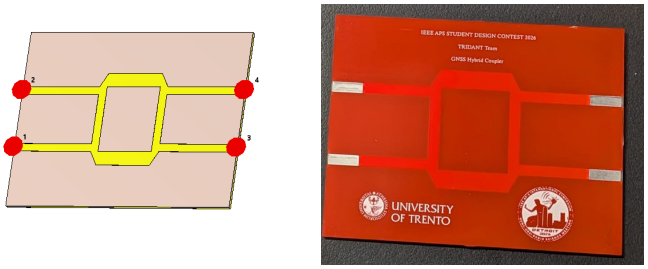
(b) Geometry of the conical air cavity employed to tune the V2X operating band.

Fig. 1: Geometry of the proposed dual-band dielectric resonator antenna and detailed view of the conical air-cavity configuration.

- O2: Realize quasi-omnidirectional horizon-oriented radiation in the V2X band.
- O3: Achieve both behaviors through modal excitation rather than geometry switching.
- O4: Develop a compact, low-cost, classroom-reproducible demonstration platform enabling real-time visualization of antenna behavior.

II. Antenna Design and Validation

The proposed antenna is a dual-band dielectric resonator antenna (DRA) designed to provide circularly polarized (CP) broadside radiation in the GNSS L1 band at 1.575 GHz and a quasi-omnidirectional radiation pattern with horizon coverage



(a) CST model of the proposed coupler.

(b) Photograph of the fabricated prototype.

Fig. 2: Geometry and fabricated prototype of the proposed hybrid branch-line coupler.

in the V2X band (5.85–5.95 GHz). The antenna employs a single cylindrical dielectric resonator whose preliminary dimensions are determined using the closed-form design equations reported by Petosa [1], enabling accurate control of the excited resonant modes and their frequency separation. At the GNSS L1 band, broadside radiation is achieved by exciting the $\mathrm{HEM}_{11\delta}$ mode using two coaxial probes symmetrically positioned along the resonator periphery. Circular polarization is obtained by feeding the two ports with a 90° phase difference. To realize the required quadrature excitation, a custom hybrid branch-line coupler was designed and fabricated. Fig. 2 shows the CST model of the proposed coupler together with the photograph of the fabricated prototype.

At the higher-frequency band, the $\mathrm{TM}_{11\delta}$ mode is excited through a centrally located coaxial probe. This mode exhibits a monopole-like field distribution, producing a quasi-omnidirectional radiation pattern suitable for vehicular communication applications. In order to properly shift the resonant frequency of the $\mathrm{TM}_{11\delta}$ mode toward the V2X center frequency at 5.9 GHz, an air cavity was introduced beneath the dielectric resonator, following the studies reported in [2]– [4]. In particular, a conical air cavity was adopted and optimized through a parametric analysis to determine the appropriate geometrical dimensions. A sketch of the proposed cavity configuration is shown in Fig. 1b.

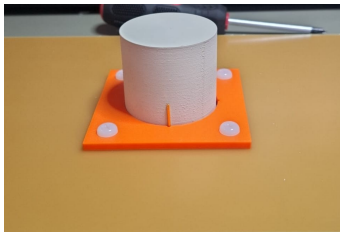
The dielectric resonator was fabricated using Zetamix material from Nanoe, characterized by a relative permittivity of approximately $\varepsilon_r \approx 7$ and a dielectric loss tangent of $\tan\delta \approx 10^{-3}$. The resonator is mounted on an FR4 substrate to provide mechanical robustness and ease of integration. Furthermore, a dedicated PLA mounting flange was designed and fabricated using 3D-printing technology to improve the mechanical stability of the overall structure.

The corresponding simulated and measured reflection coefficients for the GNSS and V2X operating bands are reported in Figs. 4a–4c, whereas the associated radiation patterns are shown in Figs. 5a–5c.

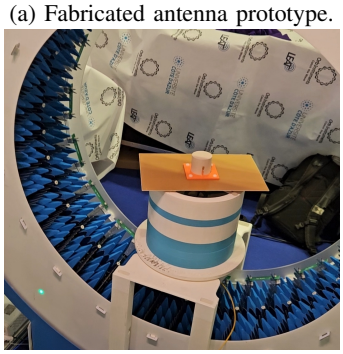
III. Experimental Demonstration Setup

A. Setup Description

To validate the radiation performance of the antenna under test, a portable, low-cost radiation pattern measurement system



(a) Fabricated antenna prototype.



(b) Measurement setup of the fabricated prototype.

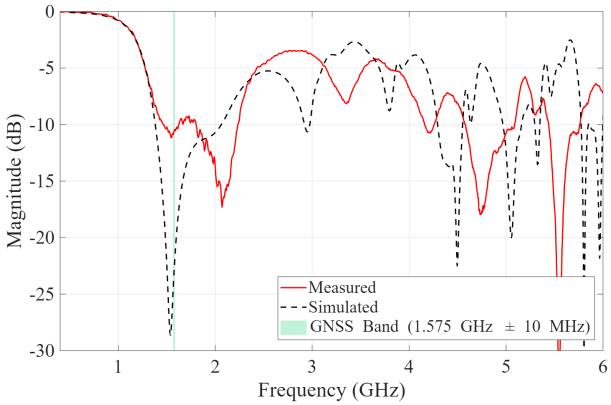
Fig. 3: Photographs of the fabricated antenna prototype and experimental characterization setup.

was developed. The transmission subsystem comprises an HT008 signal generator operating across both the Global Navigation Satellite System (GNSS) and Vehicle-to-Everything (V2X) frequency bands. This source excites a cylindrical dielectric resonator antenna (CDRA). Depending on the hardware configuration of the orthogonal feeding points, the CDRA is selectively driven to radiate either a horizontally or vertically polarized directional pattern.

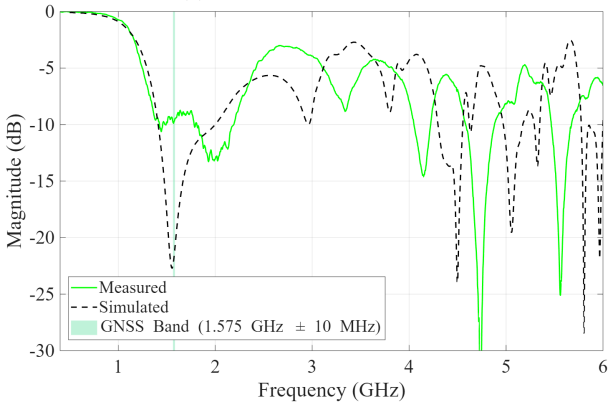
On the receiver side, the radiated field is captured via an HTOOL HT8 log-periodic directional antenna. The received RF signal is routed directly to an Analog Devices AD8317 logarithmic power detector, which provides a DC output voltage proportional to the incident RF power. This analog voltage is sampled by the internal analog-to-digital converter (ADC) of a CodeCell C6 microcontroller platform. Concurrently, an embedded BNO086 inertial measurement unit (IMU) tracks the spatial orientation and angular sweep of the receiver antenna. This allows for the real-time correlation of angular position data with the corresponding received RF power metrics.

To prevent co-site electromagnetic interference (EMI) and receiver desensitization, a time-domain isolation protocol is implemented for data telemetry. The onboard 2.4 GHz Bluetooth Low Energy (BLE) transmitter operates intermittently; the BLE radio is systematically powered down during the ADC sampling window of the RF power detector. Once the RF measurement is registered, the BLE link is re-established to transmit the synchronized orientation and power data packets to a host terminal for graphical visualization.

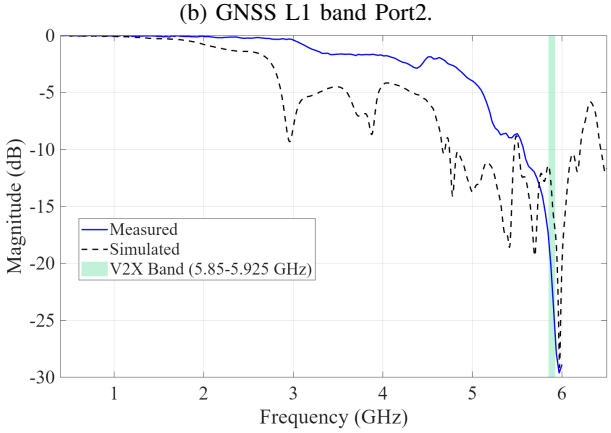
Mechanical portability is achieved by integrating the receiver circuitry, power distribution, and user controls within a custom handheld enclosure, as illustrated in Fig. 7. The system is energized by a lithium-ion battery managed via a



(a) GNSS L1 band Port1.



(b) GNSS L1 band Port2.

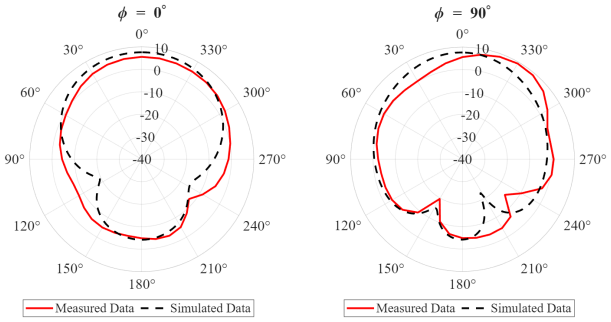


(c) V2X operating band.

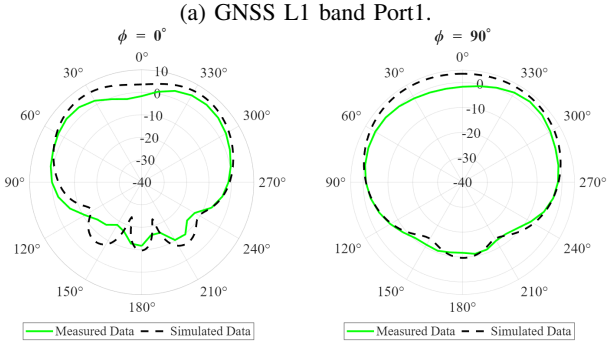
Fig. 4: Simulated and measured reflection coefficients of the proposed antenna for the GNSS and V2X operating bands.

step-up DC-DC converter module that delivers a regulated 5 V rail to power both the AD8317 detector and the CodeCell C6 platform.

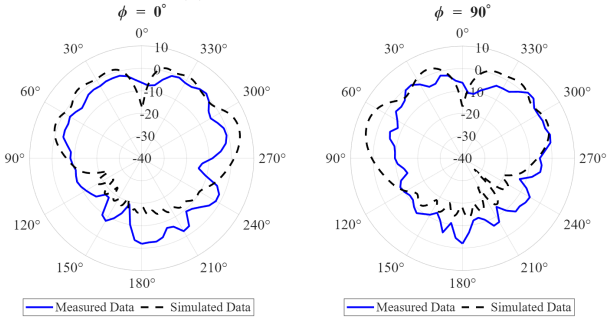
For independent spectral verification and monitoring, an auxiliary software-defined radio (SDR) platform—consisting of a Xilinx Zynq-7020 System-on-Chip (SoC) and an Analog Devices AD9361 RF agile transceiver—is connected to the directional antenna. This parallel validation setup provides high-resolution spectrum analysis, clearly isolating the effects of polarization mismatch and cross-polarization discrimination when the quadrature hybrid coupler is omitted from the CDRA feed network.



(a) GNSS L1 band Port1.



(b) GNSS L1 band Port2.



(c) V2X operating band.

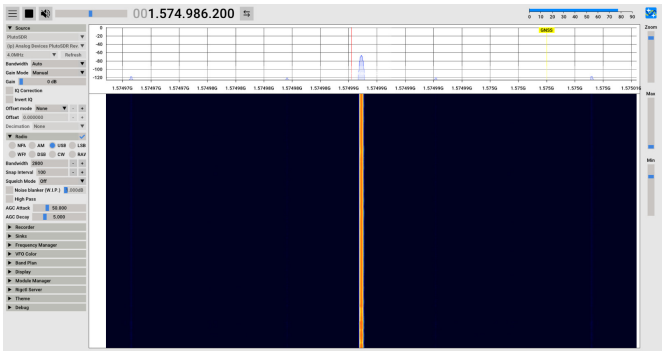
Fig. 5: Comparison between simulated and measured radiation patterns for the $\phi = 0°$ and $\phi = 90°$ cuts. For the GNSS bands, the two ports correspond to the θ- and ϕ-polarized components, respectively.

B. Parts, Materials, and Budget Analysis

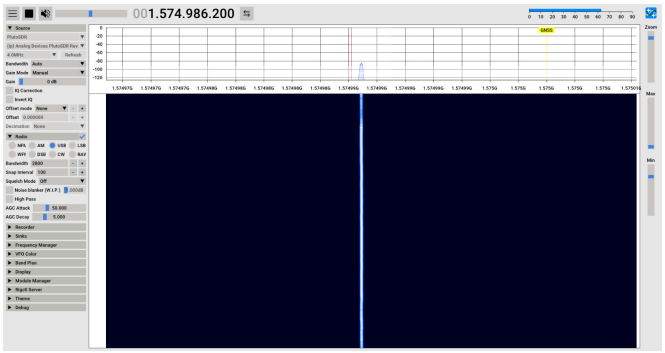
C. Physical Realization and Demonstration Layout

At the current stage of development, the real-time radiation pattern acquisition method based on the AD8317 logarithmic power detector is still undergoing software debugging and calibration. Therefore, the preliminary experimental validation of the proposed antenna characterization platform has been performed using the SDR-based measurement approach. In this configuration, the signal radiated by the HT008 source antenna was monitored through the SDR receiver while the directional probe antenna was manually rotated, as shown in Fig. 8.

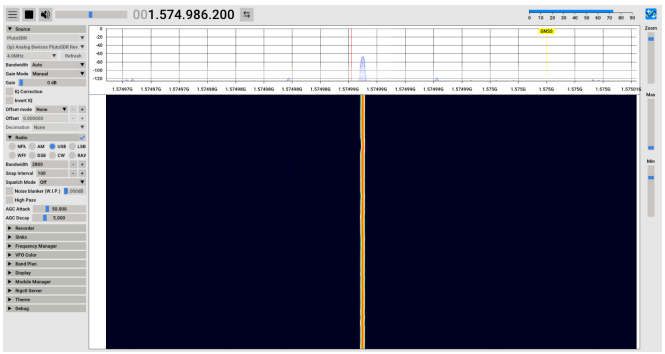
The measurements confirm the expected polarization behavior of the proposed antenna system. Specifically, when the receiving antenna is aligned with the transmitted linear polarization, the received signal level observed on the SDR spectrum display reaches its maximum value. Conversely,



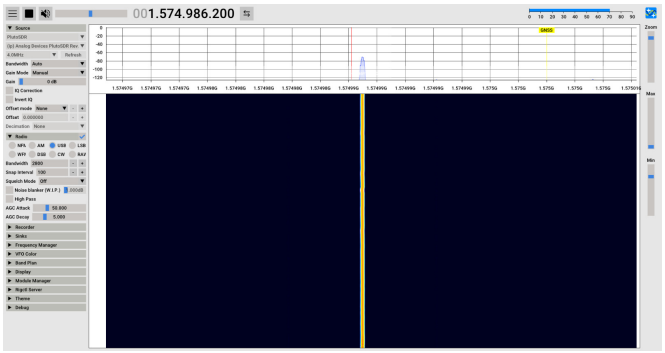
(a) Received spectrum for the co-polarized configuration without quadrature hybrid coupler.



(b) Received spectrum for the cross-polarized configuration without quadrature hybrid coupler, showing significant signal attenuation.



(c) Received spectrum with quadrature hybrid coupler integrated into the CDRA feed network for one receiving antenna orientation.

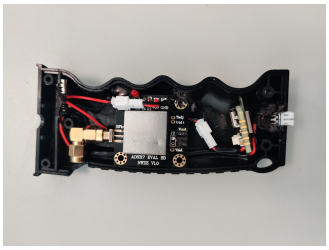


(d) Received spectrum with quadrature hybrid coupler integrated into the CDRA feed network for the orthogonal receiving antenna orientation, exhibiting nearly unchanged received power.

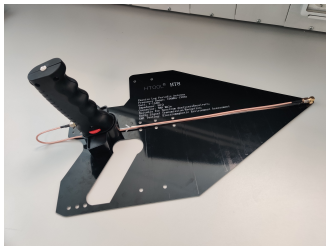
Fig. 6: SDR-based experimental verification of the polarization characteristics of the proposed antenna system. The displayed spectrum traces and waterfall plots illustrate the variation of the received signal intensity for different receiving antenna orientations and feeding configurations.

TABLE I: Bill of Materials and Cost Estimation

Item / Part	Supplier / Source	Quantity	Cost (EUR)
HT008 MAX2871	HTOOL	1	50 €
CodeCell C6	Microbots	1	30 €
AD8317 LPD	AliExpress	1	18 €
HTOOL HT8	HTOOL	1	50 €
SDR Zynq7020-AD9361	OpenSourceSDRLab	1	150€
LiPo Battery-Charging Module	AliExpress	1	2€
Nanoe Zetamix-Epsilon filament	Nanoe Zetamix	1	175€
3.7V 3000mAh-Lipo Battery	AliExpress	5	25€
SMA to SMA-coax cable	Amazon	4	25€
Hybrid Coupler	PCBWay	10	5€
Total			530€



(a) Internal component layout.



(b) Assembled field prototype.

Fig. 7: Mechanical design and hardware integration of the handheld measurement apparatus: (a) internal component layout within the handle housing, displaying the AD8317 logarithmic detector and power management routing; (b) complete field-ready assembly coupled to the HT8 log-periodic directional antenna.

when the receiving antenna is rotated to an orthogonal orientation, a significant attenuation of the received power is observed, thereby demonstrating the correct discrimination between co-polarized and cross-polarized field components.

Furthermore, when the quadrature hybrid coupler is incorporated into the CDRA feeding network, the measured received power remains substantially unchanged for different orientations of the receiving antenna. This behavior confirms the generation of circular polarization and validates the proper operation of the dual-feed excitation scheme. Representative SDR screenshots corresponding to the different polarization configurations are reported in Fig. 6, where both the spectrum trace and the waterfall visualization clearly highlight the variation in received signal intensity as a function of antenna orientation.



(a) Measurement configuration with the receiving log-periodic antenna aligned with the transmitted polarization, corresponding to the co-polarized condition.



(b) Measurement configuration with the receiving antenna rotated by 90°, corresponding to the cross-polarized condition.

Fig. 8: Experimental SDR-based validation setup used to assess the polarization properties of the proposed antenna system under different receiving antenna orientations.

IV. Conclusion and Future Developments

A dual-band dielectric resonator antenna for GNSS and V2X applications has been presented together with a compact low-cost measurement platform for radiation pattern characterization. Experimental SDR-based measurements successfully validated the expected polarization behavior, confirming both linear and circular polarization operation depending on the feeding configuration.

Future work will focus on completing the debugging and integration of the AD8317-based acquisition system to enable a fully operational live demonstration at the conference in July. Moreover, a redesigned PCB with improved geometrical symmetry will be fabricated to reduce the discrepancies observed between the two GNSS radiation patterns. Finally, an additional SDR platform will be developed to extend the experimental validation to the V2X frequency band and allow comprehensive spectral analysis across both operating bands.

Acknowledgment

The authors would like to thank the FabLab of the University of Trento for their valuable support during the fabrication and 3D-printing processes of the antenna prototypes and measurement fixtures. The authors also gratefully acknowledge Orb Instruments for providing support and facilities for the antenna measurement activities.

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[3] R. M. Baghaee, M. H. Neshati, and J. R. Mohassel, "The effects of air gap and finite ground plane on radiation characteristics of probe-fed rectangular dielectric resonator antennas using method of moment," in *Proceedings of the IEEE International Conference on Microwave and Millimeter Wave Technology (ICMMT)*, 2006, pp. 411–414.

[4] G. P. Junker, A. A. Kishk, A. W. Glisson, and D. Kajfez, "Effect of fabrication imperfections for ground-plane-backed dielectric-resonator antennas," *IEEE Antennas and Propagation Magazine*, vol. 37, no. 1, pp. 40–47, Feb. 1995.



Nicolò Cellana is currently pursuing the B.S. degree in Computer, Communications and Electronics Engineering (ICE) at the University of Trento, Italy.

His current research and academic interests include radio frequency (RF) engineering, high-frequency electronic circuits, radar systems, and optoelectronics. This project serves as the foundation for his undergraduate thesis, focusing on the design, physical integration, and software-defined radio (SDR)-based experimental validation of real-time antenna characterization platforms.



Vittoria Zandonà is currently pursuing the B.S. degree in Computer, Communications and Electronics Engineering (ICE) at the University of Trento, Italy. Her technical interests lie at the intersection of applied physics and electronics engineering, with a particular focus on analog and digital design, photonics, and quantum hardware. Within the Student Design Contest, she focused mainly on embedded systems architecture and low-level programming, telemetry and measurement-system optimization for the experimental setup.



Leonardo Di Vincenzo received the bachelor's and master's degrees in electronic engineering. He is currently pursuing the Ph.D. degree in the National PhD Program in Space Science and Technology at the University of Trento. His research focuses on the electromagnetic modeling and characterization of absorbers for anechoic chambers used in space testing applications. His interests include computational electromagnetics, microwave engineering, antenna measurements, electromagnetic compatibility, and numerical methods for high-frequency systems. He is involved in research activities related to electromagnetic simulation techniques and advanced testing environments for aerospace and satellite applications, with emphasis on accurate material modeling and experimental validation methodologies.



Leonardo Lizzi (Senior Member, IEEE) is an Associate Professor at the University of Trento, Italy. Until late 2023, he was a Maître de Conférences at Université Côte d'Azur (France), where he also completed a post-doctoral fellowship from 2011 to 2014. He received his M.Sc. and Ph.D. degrees from the University of Trento, with research visits at Penn State (USA) and Nagasaki University (Japan). Co-author of over 160 papers, his current research focuses on reconfigurable and compact antennas for IoT and space applications, and electromagnetic modeling.